

Act II: Casting the Buyers

Motivations and Choices of the Citizens

Module III: Citizen Motivations

Consumer Choice, Utility, and Budgets



Previously in Marginal Falls

- ◆ Scarcity
- ◆ Opportunity Cost
- ◆ Marginal Decisions
- ◆ Comparative Advantage
- ◆ Gains from Trade

Next: What do citizens want?

How Do Citizens Choose?

Why does one citizen buy coffee, another buy a pastry, and another save the money?



What do citizens hope to get when they consume something?



Utility

Utility is the satisfaction a person receives from consuming a good or service.

- Coffee, pastries, ferry rides, clues, and festival tickets can all give utility.

Marginal Utility is the change in total utility a person receives when consuming one additional unit of a good or .

The Marginal Falls Murder

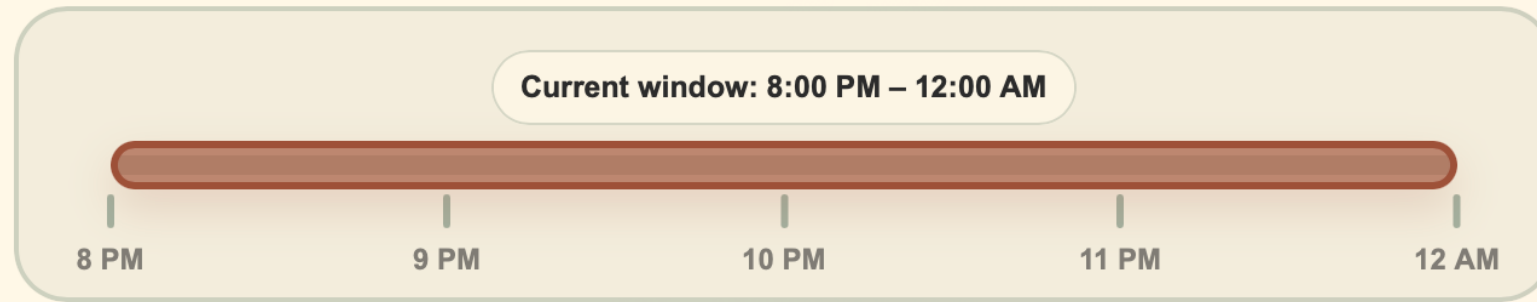
<https://econgames.promitkchaudhuri.com/murder/>

- Buy clues.
- Narrow the time window.
- Guess the time of death.
- Maximize your score.



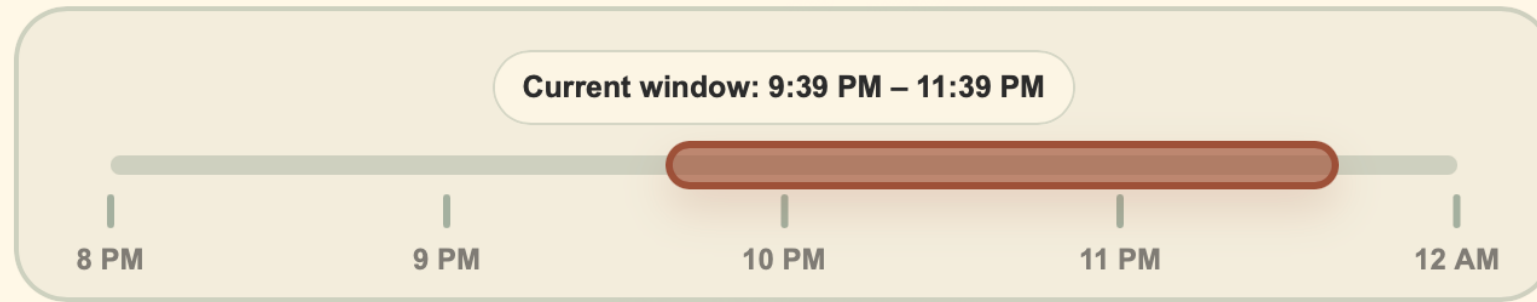
Is one more clue worth it?

What did the clues do?



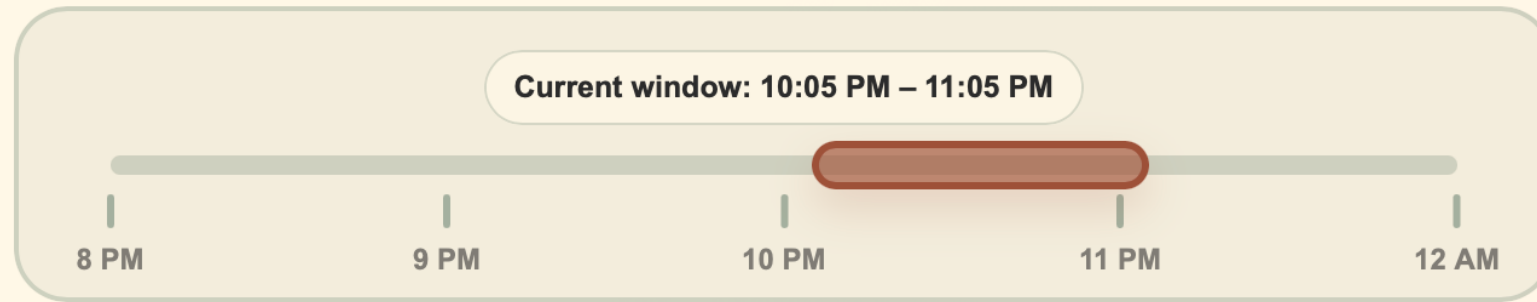
What the first clue buys you?

What did the clues do?



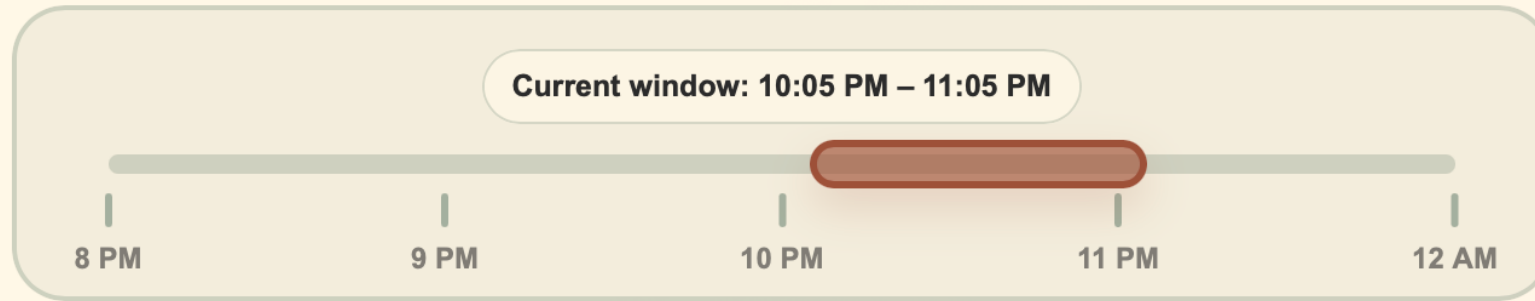
What the second clue buys you?

What did the clues do?



Was the second clue equally useful as the first clue?

What did the clues do?



Was the second clue equally useful as the first clue?

The marginal utility of a clue is the extra usefulness of buying one more clue.

Marginal Utility of a clue decreases as the number of clue increases.

The Principle of Diminishing Marginal Utility

Law of Diminishing marginal utility

The principle that each additional unit often gives less extra satisfaction than the previous one.

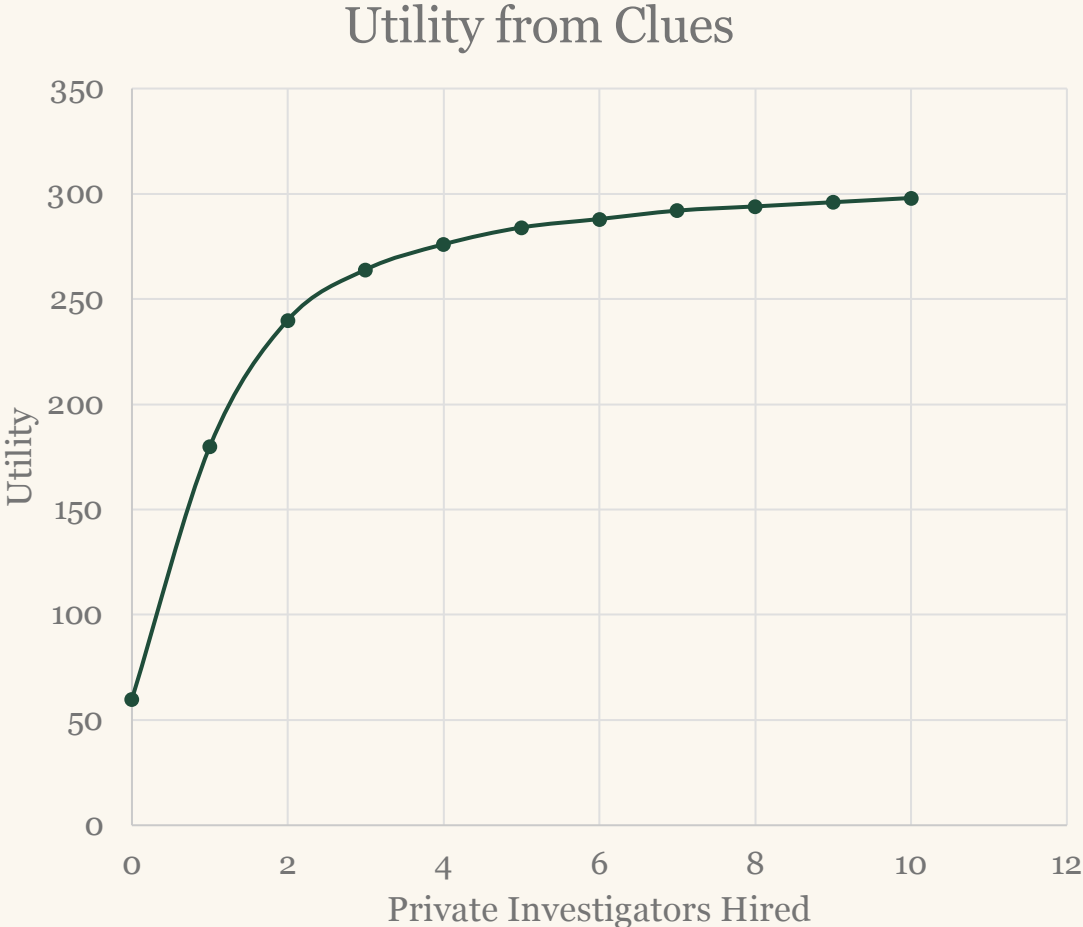
Compare the satisfaction you get from eating pizza slices:

Hungry → First Slice

Third Slice → Fourth Slice

Clues and Utility

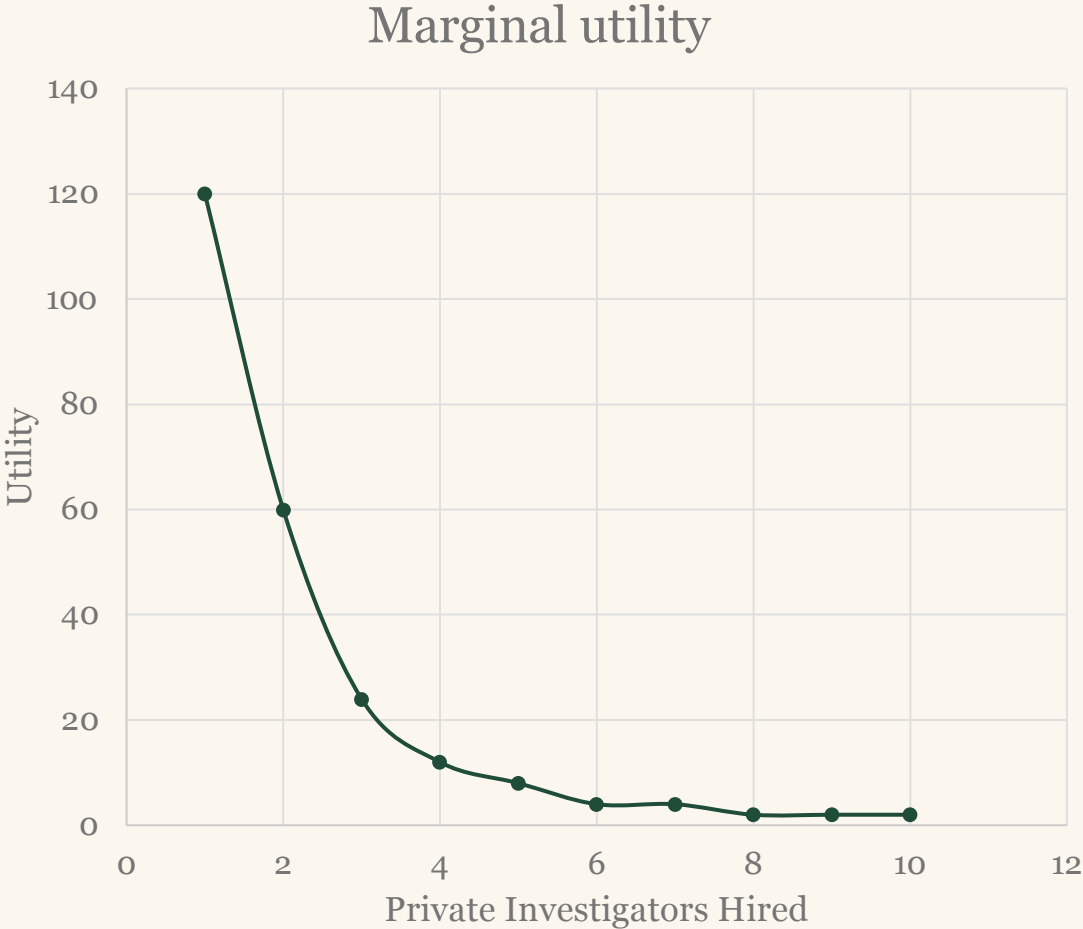
Utility = 300 – Interval



PIs	Interval (in mins)	Utility (in utils)
0	240	60
1	120	180
2	60	240
3	36	264
4	24	276
5	16	284
6	12	288
7	8	292
8	6	294
9	4	296
10	2	298

Clues and Utility

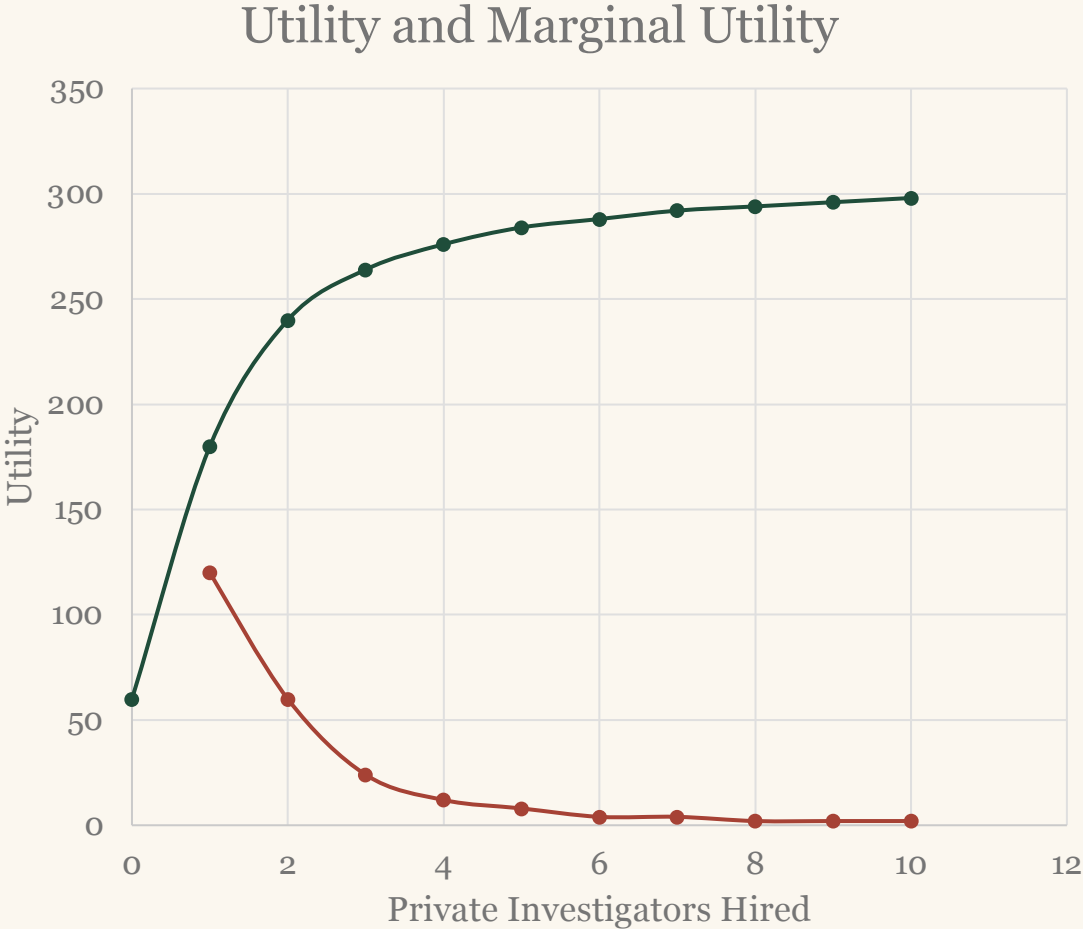
Utility = 300 – Interval



PIs	Interval (in mins)	Utility (in utils)	Marginal Utility
0	240	60	-
1	120	180	120
2	60	240	60
3	36	264	24
4	24	276	12
5	16	284	8
6	12	288	4
7	8	292	4
8	6	294	2
9	4	296	2
10	2	298	2

Clues and Utility

Utility = 300 – Interval



PIs	Interval (in mins)	Utility (in utils)	Marginal Utility
0	240	60	-
1	120	180	120
2	60	240	60
3	36	264	24
4	24	276	12
5	16	284	8
6	12	288	4
7	8	292	4
8	6	294	2
9	4	296	2
10	2	298	2

The Buyer's Constraint

Until now: what citizens want.

Way Ahead: what citizens can afford.

Choices require both preferences and constraints.

Utility Alone is not Enough

Citizens want many things.

But they cannot buy everything.



Scarcity returns.

Budget Constraint

The limited amount of income available to consumers to spend on goods and services.

Marginal Falls: A citizen may want everything in town, but the wallet limits the choice.

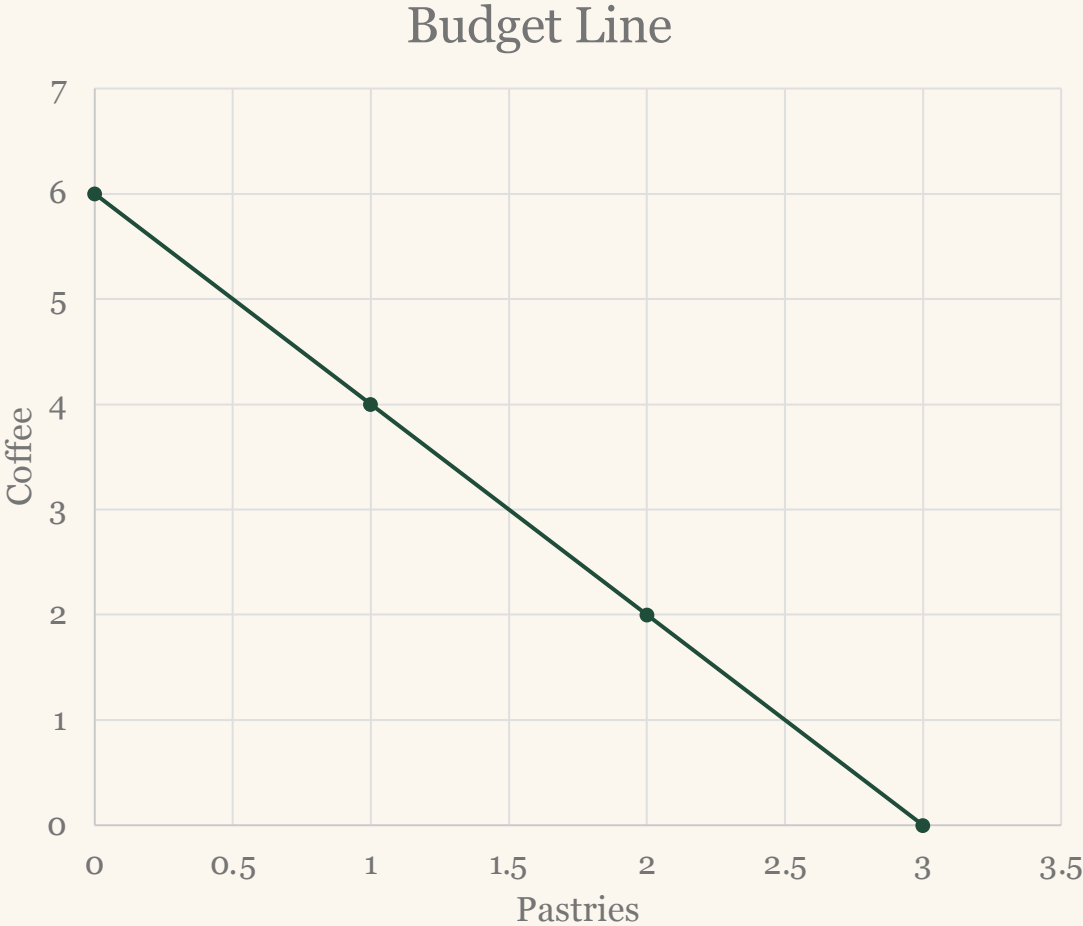
What can the citizen buy?

Assume:

- Budget = \$12
- Coffee = \$2
- Pastry = \$4

Pastries	Coffee	Spending
0	6	\$12
1	4	\$12
2	2	\$12
3	0	\$12

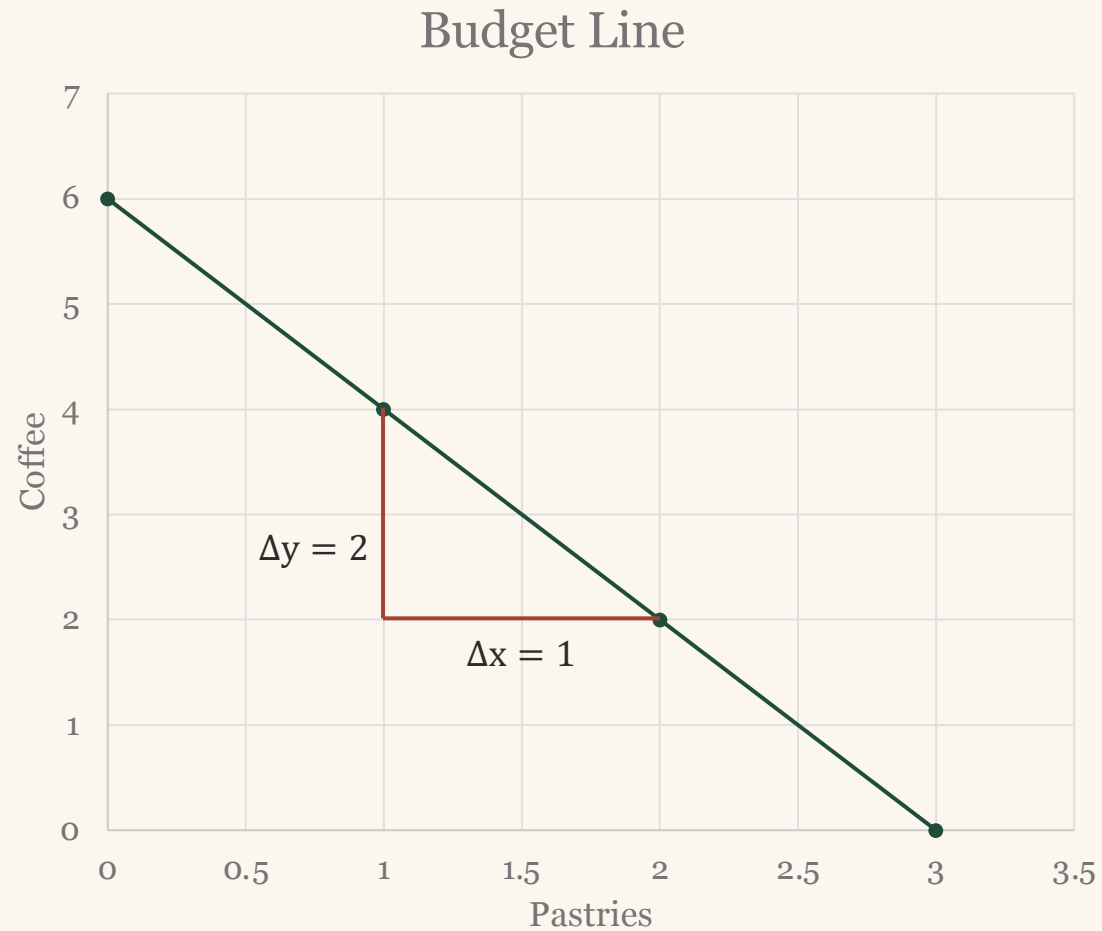
Budget Line



Pastries	Coffee	Spending
0	6	\$12
1	4	\$12
2	2	\$12
3	0	\$12

The **budget line** shows all combinations that exactly spend the budget.

Budget Line is a Trade-Off



Pastries	Coffee	Spending
0	6	\$12
1	4	\$12
2	2	\$12
3	0	\$12

To get 1 more pastry, the citizen gives up 2 coffees.

Back to the Murder Case

- The time of death has been narrowed down.
- But the police still need enough evidence to convict.
- The Chief now needs to allocate budget for the case.



How should the investigation budget be spent?

Operation: Contingency Plan

<https://eongames.promitkchaudhuri.com/detective>

The Chief gives you \$10,000.

You can spend it on: **Detectives or Forensics**

Cost: \$1,000 each

Goal: Build the strongest possible case without going over budget.



What Did You Choose?

- Did you spend the whole budget?
- Did you buy mostly detectives?
- Did you buy mostly forensics?

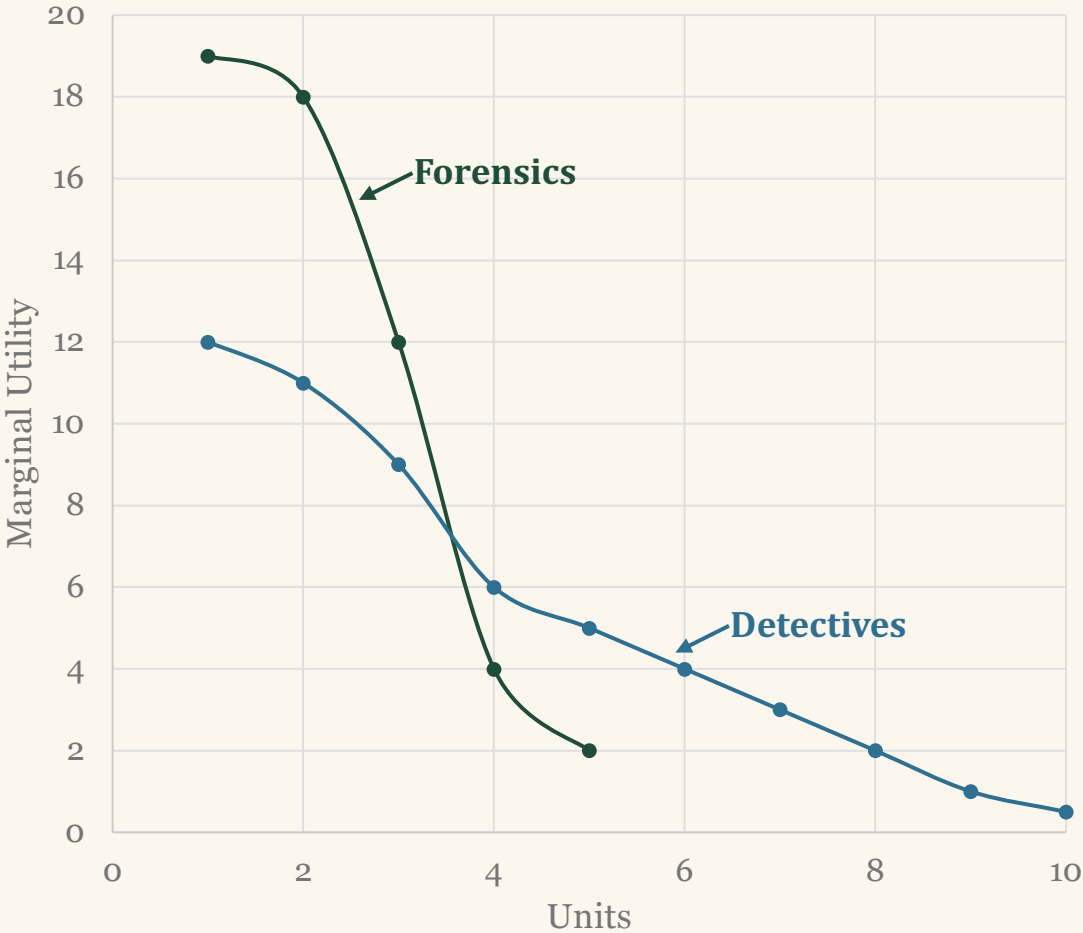
The Conviction Table

Unit	Conviction Rate from Forensics	Extra Gain	Conviction Rate from Detectives	Extra Gain
0	0	-	0	-
1	19	19	12	12
2	37	18	23	11
3	49	12	32	9
4	53	4	38	6
5	55	2	43	5
6	-	-	47	4
7	-	-	50	3
8	-	-	52	2
9	-	-	53	1
10	-	-	53.5	0.5

The Conviction Table

Unit	Total Utility from Forensics	Marginal Utility ($MU_{Forensics}$)	Total Utility from Detectives	Marginal Utility ($MU_{Detectives}$)
0	0	-	0	-
1	19	19	12	12
2	37	18	23	11
3	49	12	32	9
4	53	4	38	6
5	55	2	43	5
6	-	-	47	4
7	-	-	50	3
8	-	-	52	2
9	-	-	53	1
10	-	-	53.5	0.5

Diminishing Returns in the Case File



Unit	Total Utility from Forensics	Marginal Utility ($MU_{Forensics}$)	Total Utility from Detectives	Marginal Utility ($MU_{Detectives}$)
0	0	-	0	-
1	19	19	12	12
2	37	18	23	11
3	49	12	32	9
4	53	4	38	6
5	55	2	43	5
6	-	-	47	4
7	-	-	50	3
8	-	-	52	2
9	-	-	53	1
10	-	-	53.5	0.5

Marginal Utility per Dollar Spent

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

Marginal Utility per dollar spent:

Marginal Utility of Forensics divided by the Price of Forensics, i.e. \$1000.

Similarly, for Detectives.

Conviction gain per dollar spent.

The Buyer's Rule

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

The Buyer's Rule

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

The Buyer's Rule

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

- Spend the first \$1000 dollar on Forensics:
 $0.019 > 0.012$

The Buyer's Rule

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

- Spend the first \$1000 dollar on Forensics:
 $0.019 > 0.012$
- Choose the next unit that gives the highest marginal utility per dollar.
- At the best bundle:
Marginal utility per dollar spent is equal.

Equalize the “bang for the buck”.

Equalizing Marginal utility per dollar spent

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

Forensics	Detectives	MU/P	Total Spending

Equalizing Marginal utility per dollar spent

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

Forensics	Detectives	MU/P	Total Spending
3	1	0.012	\$4000

Equalizing Marginal utility per dollar spent

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

Forensics	Detectives	MU/P	Total Spending
3	1	0.012	\$4000
4	6	0.004	\$10,000

Equalizing Marginal utility per dollar spent

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.019	12	0.012
2	18	0.018	11	0.011
3	12	0.012	9	0.009
4	4	0.004	6	0.006
5	2	0.002	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

Forensics	Detectives	MU/P	Total Spending
3	1	0.012	\$4000
4	6	0.004	\$10,000
5	8	0.002	\$13,000

Budget Constraint

Forensics	Detectives	MU/P	Total Spending
3	1	0.012	\$4000
4	6	0.004	\$10,000
5	8	0.002	\$13,000

The actual combination depends on the budget constraint.

Budget Constraint

Forensics	Detectives	MU/P	Total Spending	
3	1	0.012	\$4000	✗
4	6	0.004	\$1000	✓
5	8	0.002	\$13000	✗

Optimal Bundle:
(4 *Forensics*, 6 *Detecivtes*)

Optimal Choice

Two conditions for maximizing utility:

- Satisfy the rule of equal marginal utility per dollar spent:

$$\frac{MU_F}{P_F} = \frac{MU_D}{P_D}$$

- Exhaust your budget:

Spending on Forensics + Spending on Detectives = Budget

Same Satisfaction, Different Bundles

Suppose a citizen likes both coffee and pastries.

They may be equally satisfied with:

- 2 coffees and 2 pastries
- 4 coffees and 1 pastry

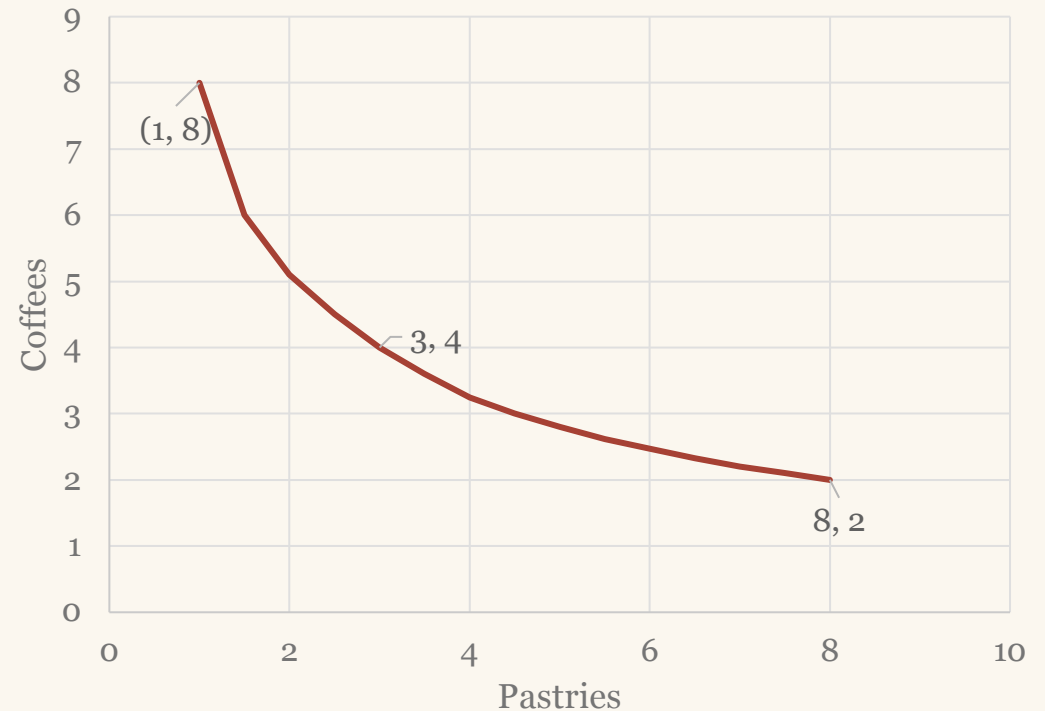
An **indifference curve** shows bundles that give the consumer the same utility.

Same Satisfaction, Different Bundles

Suppose a citizen likes both coffee and pastries.

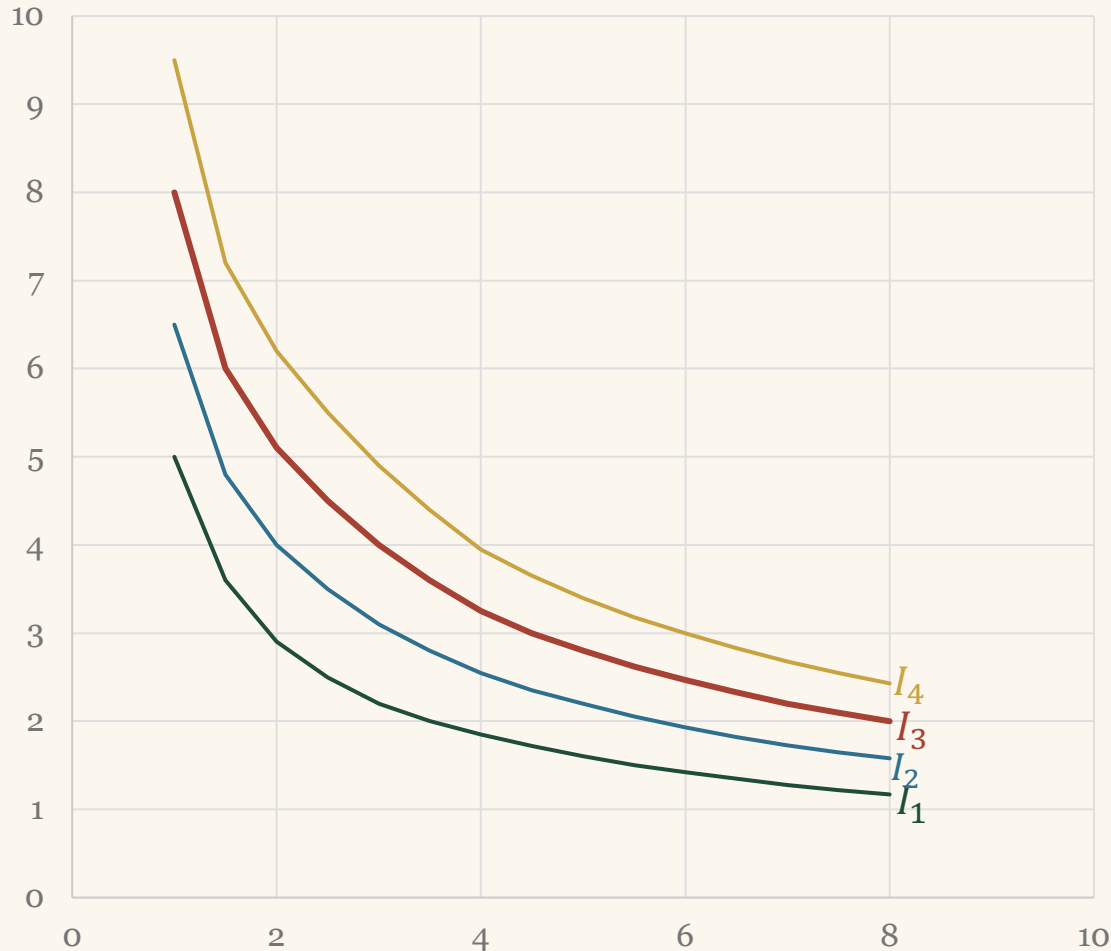
They may be equally satisfied with:

- 4 coffees and 3 pastries
- 8 coffees and 1 pastry



An **indifference curve** shows bundles that give the consumer the same utility.

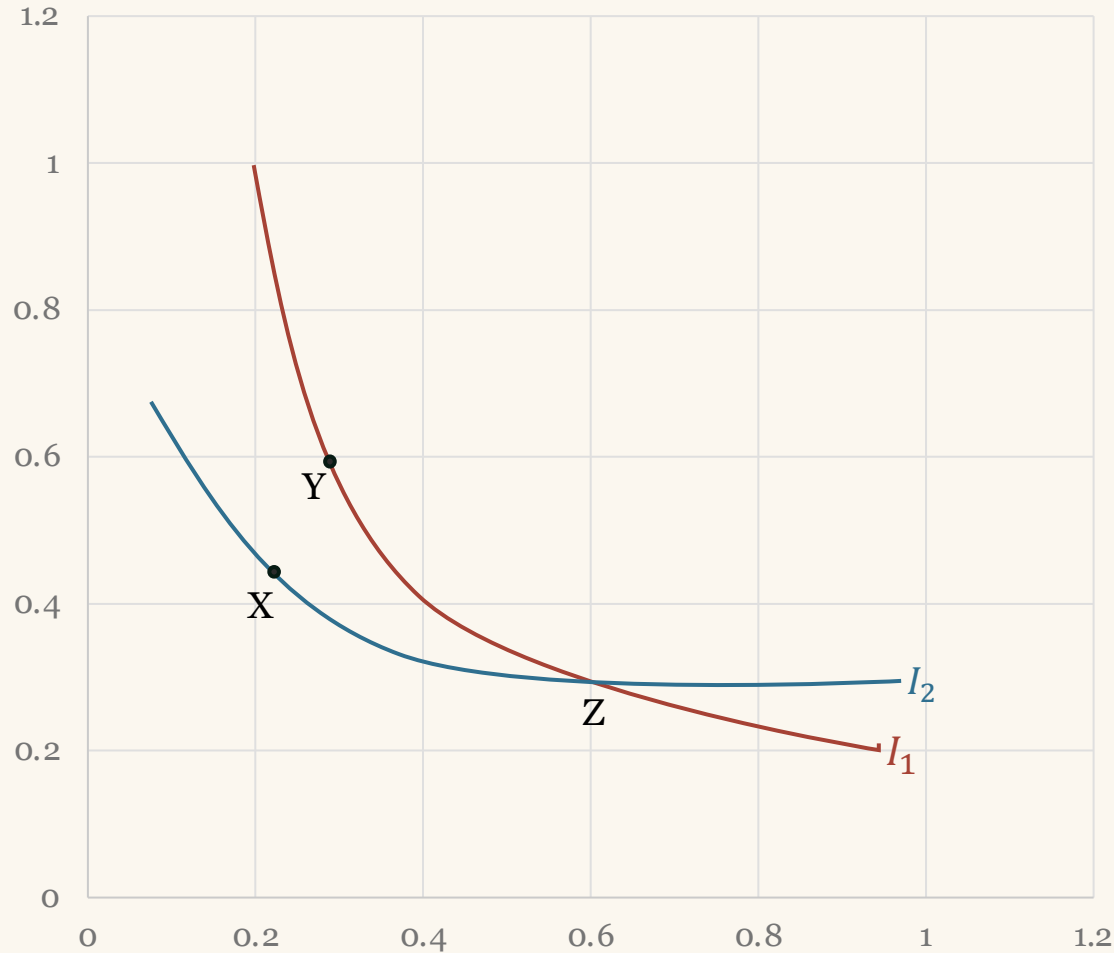
Higher Curves are Better



- Every point on one indifference curve gives the same utility.
- But not all curves are equally good.
- A higher indifference curve gives more utility.

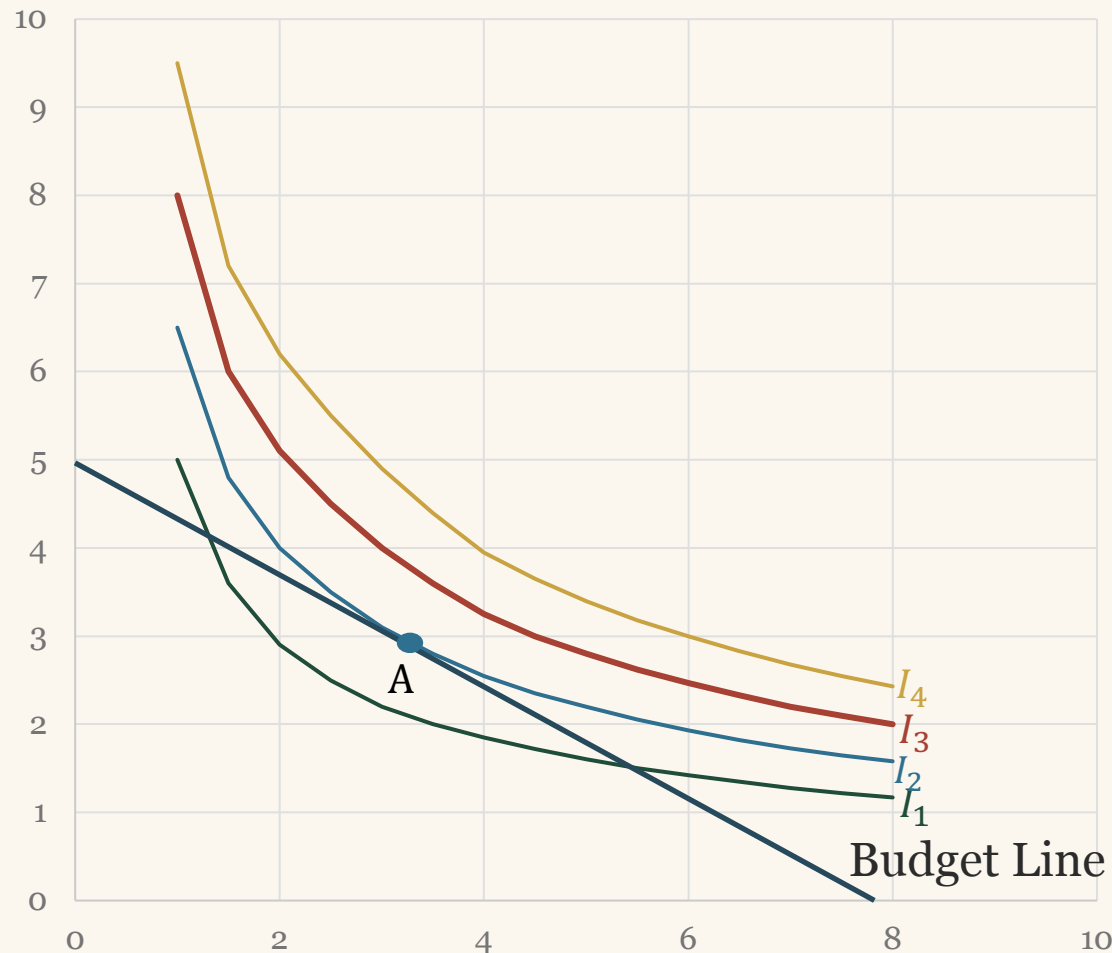
$$I_1 < I_2 < I_3 < I_4$$

Indifference Curves never Intersect



- X and Z are indifferent.
- Y and Z are indifferent.
- Then, X and Y must be indifferent as well.
- But Y is higher than X.

Preferences Meet the Budget



- Indifference curves show what citizens like.
- The budget line shows what citizens can afford.
- The chosen bundle is the best affordable bundle.

The citizen chooses the highest indifference curve they can reach.

What if the Price changes?

<https://eongames.promitkchaudhuri.com/detective>

The chief just realized he is not sure about the exact price of forensics.

He knows it is more expensive than \$1,000, but he cannot remember the exact quote.

He wants you to make contingency plans for if Forensics costs \$2,000 or \$3,000 per unit.

Two Forces Behind the New Choice

When the price Forensics increases, two things happen:

1. Substitution Effect

Forensics become more expensive relative to Detectives.

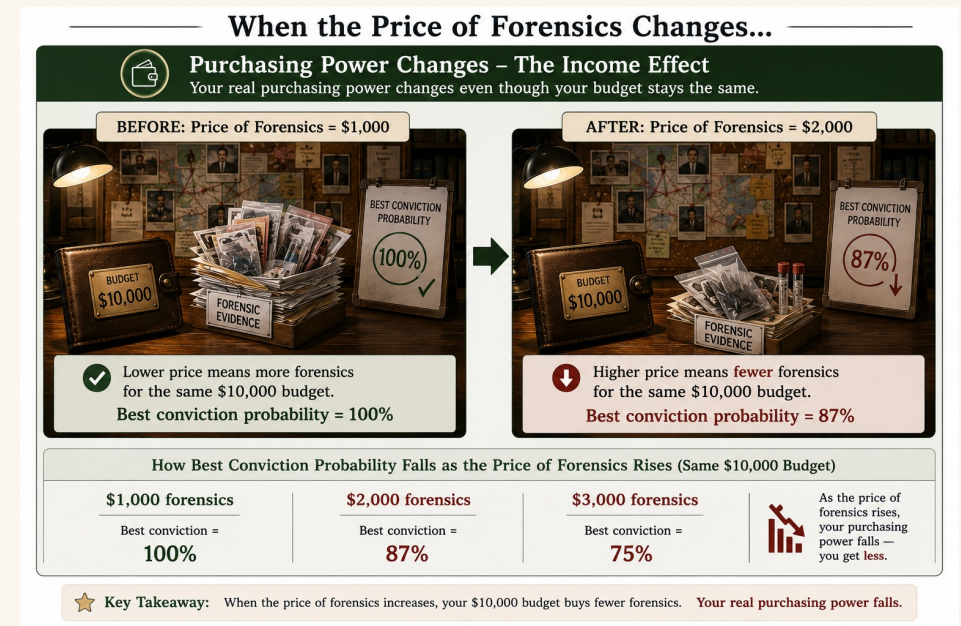
2. Income Effect

The same budget can now buy less than before.

Income Effect

Change in choice caused by a change in purchasing power.

The Chief still has \$10,000, but that money does not buy as much evidence when forensics becomes more expensive.



Income Effect

Change in choice caused by a change in purchasing power.

Normal Goods: Goods that we consume more of as our income rises

- Festival Tickets, Apartments with river views, Vacations

Inferior Goods: Goods that we consume less of as our income rises

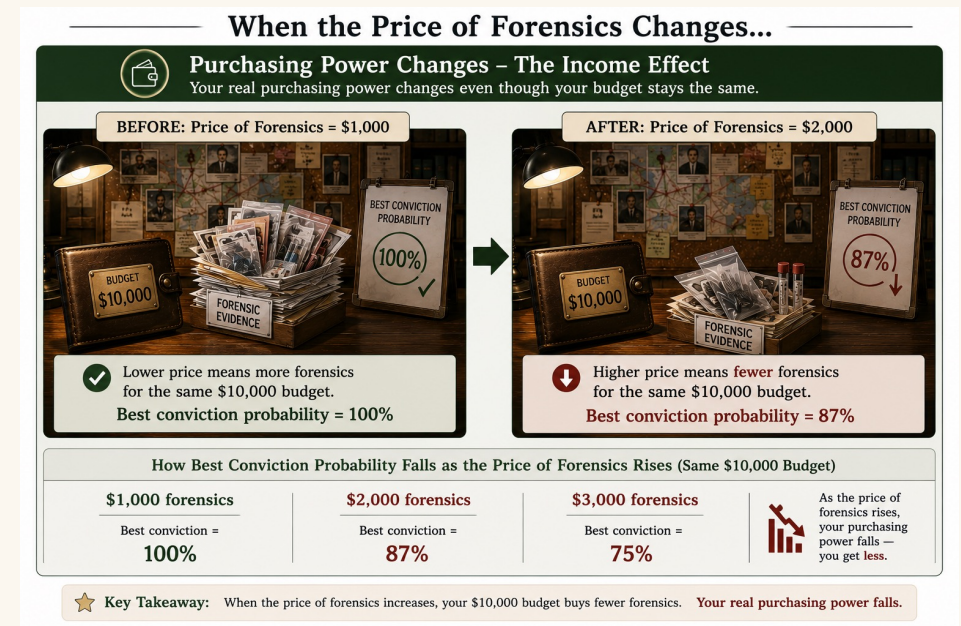
- Second-hand winter coats, Instant Noodles

- For a Normal Good, it's price decreasing $\Rightarrow$ you consume more of it $\Rightarrow$ positive IE
- For an Inferior Good, it's price decreasing $\Rightarrow$ you consume less of it $\Rightarrow$ negative IE

Substitution Effect

Change in choice caused by one good becoming more or less expensive relative to another good.

When forensics becomes more expensive, detectives become relatively cheaper.



Price Change

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.0095	12	0.012
2	18	0.009	11	0.011
3	12	0.006	9	0.009
4	4	0.002	6	0.006
5	2	0.001	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

$$P_F = \$2000$$

Forensics	Detectives	MU/P	Total Spending
2	3	0.009	\$7000
3	4	0.006	\$10,000
4	8	0.002	\$16,000
5	9	0.001	\$19,000

Price Change

Unit	Marginal Utility (Forensics) $MU_{Forensics}$	Marginal utility per dollar spent $\left(\frac{MU_{Forensics}}{P_{Forensics}}\right)$	Marginal Utility (Detectives) $MU_{Detectives}$	Marginal utility per dollar spent $\left(\frac{MU_{Detectives}}{P_{Detectives}}\right)$
0	-	-	-	
1	19	0.0095	12	0.012
2	18	0.009	11	0.011
3	12	0.006	9	0.009
4	4	0.002	6	0.006
5	2	0.001	5	0.005
6	-	-	4	0.004
7	-	-	3	0.003
8	-	-	2	0.002
9	-	-	1	0.001
10	-	-	0.5	0.0005

$$P_F = \$2000$$

Forensics	Detectives	MU/P	Total Spending
2	3	0.009	\$7000
3	4	0.006	\$10,000
4	8	0.002	\$16,000
5	9	0.001	\$19,000

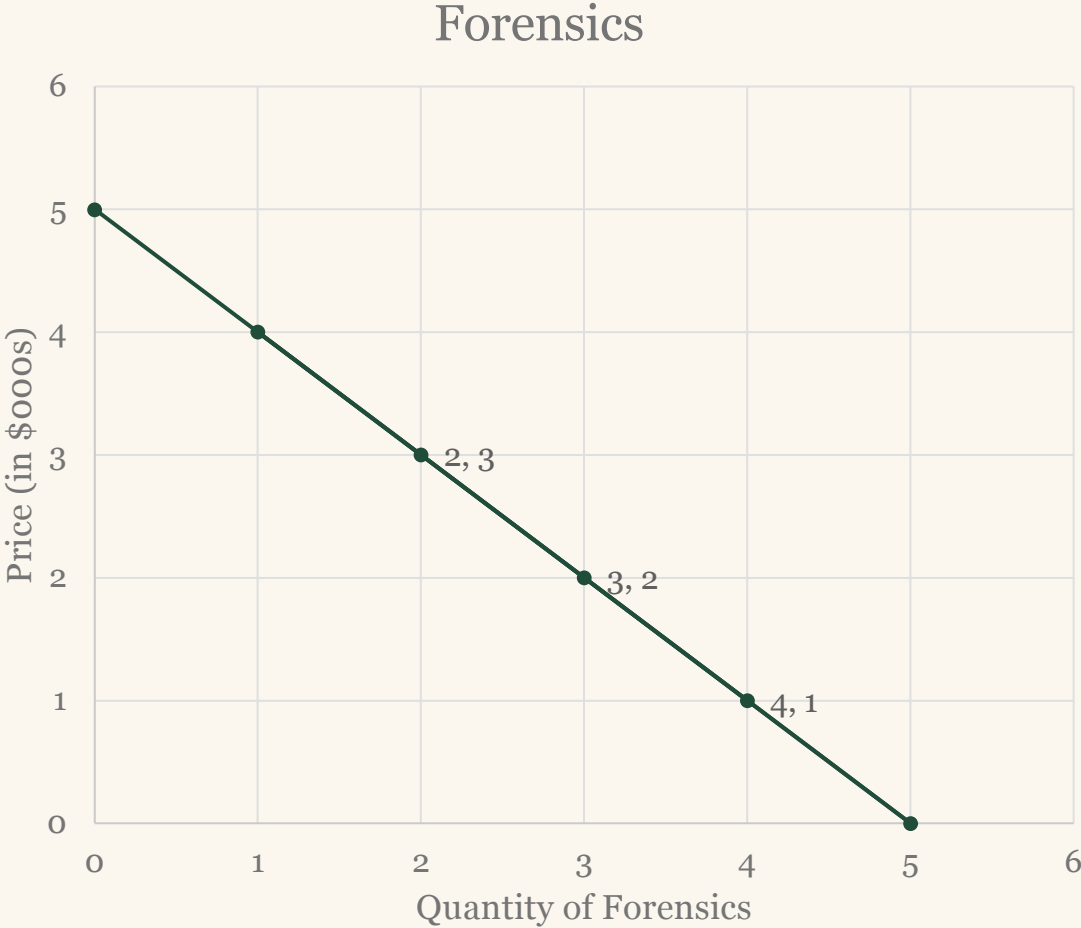
Optimal Bundle:
(3 Forensics, 4 Detectives)

Three Prices, Three Choices

Price of Forensics	Forensics	Detectives	Conviction Rate
\$1000	4	6	100%
\$2000	3	4	87%
\$3000	2	4	75%

As the price of forensics rises, the chosen quantity of forensics falls.

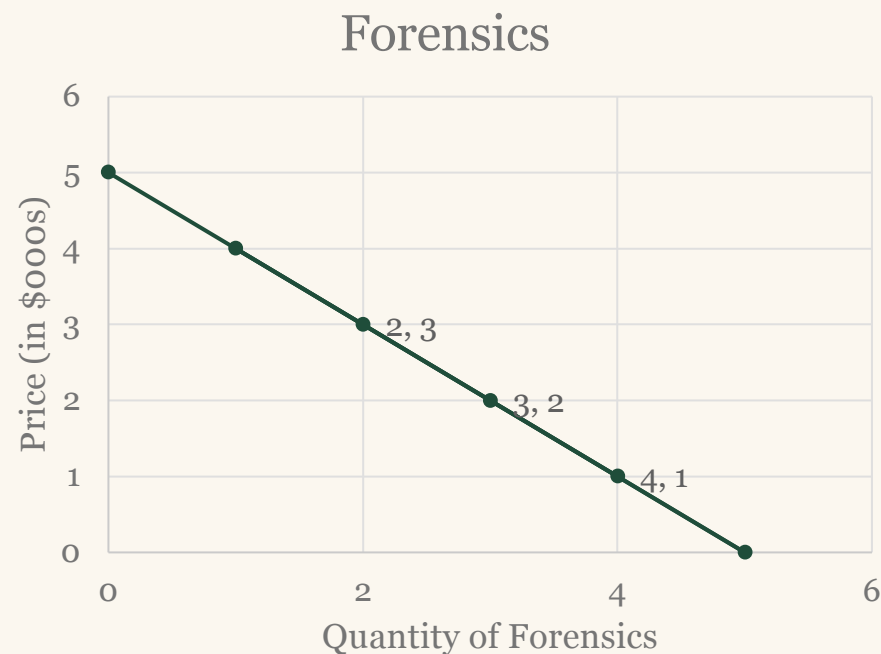
What does this look like?



Price of Forensics	Forensics	Detectives	Conviction Rate
\$1000	4	6	100%
\$2000	3	4	87%
\$3000	2	4	75%

Demand

Demand shows the relationship between the price of a good and the quantity consumers are willing and able to buy, holding other things constant.



The demand for Forensics shows how many forensic units you recommend to the Chief at each price.